Operational Audit and Strategic Architecture Report: Pointer-First Agent Constitution and Subsystem Integration for Stela

1. Executive Summary

1.1 Report Scope and Objectives

This report provides an exhaustive architectural audit and operational analysis of the "Stela" repository, explicitly focusing on the implementation and refinement of the "Pointer-First" agent constitution. This document responds to the imperative for a rigorous transition from component certification (Agents, Skills, Tasks) to a fully integrated, functional ecosystem capable of autonomous output. The analysis evaluates the current repository state following the execution of Dispatch Packets (DP) 0041, 0042, and 0043, and formulates the precise requirements for **DP-OPS-0044**.

The analysis proceeds under the strict "Behavioral Logic Standards" defined in the repository's Policy of Truth (PoT.md), necessitating linguistic precision, quantitative reporting, and the prohibition of contractions.[1] The objective is to determine whether the constituent parts of the Stela system—specifically the harvesting regimes and automated gates—are functioning with sufficient fidelity to support complex, multi-agent orchestration.

The core objectives of this operational audit are as follows:

1. **Verify Subsystem Integrity:** Assess whether the Agents, Skills, and Tasks subsystems operate within the hard constraints of PoT.md and whether the automated enforcement mechanisms (scripts, lints, tests) execute properly and on schedule.[1]
2. **Evaluate Harvesting Regimes:** Analyze the efficacy of the current "Harvesting" algorithms for agents and tasks, specifically addressing the requirement for "Fewer Drafts - More Quality" through the refinement of heuristics such as "Pattern Density" and "Registry Locking".[1]
3. **Assess Cross-Subsystem Integration:** Determine the functional capacity of the Agent-Skill-Task triad to produce tangible outputs beyond theoretical governance, ensuring that the system allows agents to effectively utilize skills to complete tasks.[1]
4. **Formulate DP-OPS-0044:** Synthesize these findings into a rigorous Dispatch Packet that locks in recent gains, refines harvesting logic to eliminate semantic duplication, and mandates the next phase of integration testing.

1.2 State of Play Assessment

The Stela repository currently occupies a state of **Active Consolidation**. The governance framework, anchored by PoT.md and enforced by the Integrator (Lead AI), has successfully established a "Flat Truth" topology where canon surfaces remain distinct from operational binaries.[1]

Recent operations have yielded the following status indicators:

- **Agents (DP-OPS-0042):** The agent subsystem has undergone hardening with "Pattern Density" heuristics to detect role emergence. However, the system currently requires manual "Harvest Check" overrides to generate reports, indicating a deficiency in autonomous background processing. Furthermore, the "Freshness Gate" required an operator override, signaling a potential drift in branch management protocols.[1]
- **Tasks (DP-OPS-0043):** The task subsystem successfully implemented "Registry Collision Locks" to prevent duplicate ID generation. The "Harvester Closeout Probe" confirmed that task templates now correctly point to TASK.md Section 4 for closeout procedures. Metrics indicate substantial refactoring, with 210 lines added to TASK.md to support this hardening.[1]
- **Skills (DP-OPS-0041):** The skills subsystem has undergone an overhaul to align with the pointer-first doctrine. However, specific metrics on skill utilization in active tasks remain sparse in the provided logs, suggesting a gap in the verification of skill ingestion.[1]

Critical Observation: While the individual components (Agents, Skills, Tasks) are certified as "serviceable," the *connective tissue*—the orchestration logic allowing an Agent to autonomously pull a JIT (Just-in-Time) Skill to execute a Task—remains heavily reliant on Operator mediation. The current scheduling of verification scripts is purely event-driven (triggered by Pull Requests or Pushes) rather than schedule-driven, leaving the system vulnerable to "Silent Drift" during periods of low operator activity.[1]

1.3 Strategic Recommendations for DP-OPS-0044

To advance Stela from a collection of certified parts to a functional production engine, DP-OPS-0044 must prioritize **Integration Density**. This requires the following strategic actions:

1. **Tightening Harvesting Algorithms:** The system must increase the threshold for "Pattern Density" in ops/lib/scripts/agent.sh to reduce low-quality drafts and implement "Semantic Locking" in ops/lib/scripts/task.sh to prevent synonymous task duplication.
2. **Enforcing Schedule-Like Rigor:** As native cron schedules are absent in .github/workflows [1], DP-OPS-0044 must codify a "Daily Open" ritual or similar operator-triggered heartbeat to ensure lints and verification scripts run regularly.
3. **Mandating Interaction Tests:** The Dispatch Packet must introduce a "Proving Ground" task that explicitly requires an Agent to load a Skill and modify a file, verifying the full

causal chain of the Pointer-First constitution.

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2. Architectural Audit: The Pointer-First Constitution

2.1 The Policy of Truth (PoT) as Physical Law

The governance of Stela is defined by PoT.md, which functions not merely as documentation but as the "Physical Law" of the repository. The central tenet is the **Filing Doctrine**, which strictly segregates the repository into functional domains to prevent cognitive contamination and structural entropy.

2.1.1 The Filing Doctrine and Jurisdiction

The Filing Doctrine separates the repository into four distinct jurisdictions, each with specific rules for content and interaction [1]:

Directory	Designation	Function	Content Rules
ops/	**Run**	Binaries, Manifests, Automation	Kinetic layer. Executable logic only. No narrative.
docs/	**Explain**	Manuals, Rationale, Governance	Semantic layer. Explanations and pointers. No code.
projects/	**Work**	Payload Code	Output layer. The actual product. Subject to PoT.
storage/	**Trash**	Artifacts, Archives, Logs	Ephemeral layer. Never Canon.

This separation ensures that the "Integrator" (the governance AI) can audit the system without parsing project-specific business logic, while the "Contractor" (the worker AI) can execute tasks without altering the fundamental laws of the system.

2.1.2 The "Pointer-First" Philosophy

The "Pointer-First" philosophy dictates that knowledge must not be duplicated. Instead of repeating instructions, the system must rely on **Pointers** (references to canonical files). This

architecture minimizes "Context Hazards"—the risk of the AI context window becoming polluted with stale or conflicting instructions.

Audit Finding: The current state of ops/lib/manifests/CONTEXT.md correctly excludes docs/library/agents, docs/library/tasks, and docs/library/skills.[1] This exclusion is critical because these library files are **JIT (Just-in-Time)** resources; they should only be loaded into the context window when specifically invoked by a Dispatch Packet. The recent logs from DP-OPS-0043 indicate that tools/lint/context.sh is passing [1], confirming that the "Pointer-First" hygiene is currently intact. However, the system relies entirely on the operator running these lints manually or via PR gates; there is no passive monitoring.

2.2 The Staffing Protocol: Authority and Logic

The integrity of the Stela system depends on a rigid **Staffing Protocol** that delineates authority between Human and AI actors. This protocol is not a suggestion but a hard constraint defined in PoT.md Section 4.1.[1]

- **Operator (Human):** The Operator holds total jurisdiction. They own final decisions, approvals, and secrets. Crucially, they are the **only** entity permitted to perform git commits, pushes, and merges. This "Human-in-the-Loop" architecture serves as the ultimate fail-safe against AI hallucination or rogue recursion.
- **Integrator (Lead AI):** The Integrator operates the governance layer. Its primary responsibilities are maintaining structural integrity, detecting "Drift" (divergence from PoT), and generating Dispatch Packets. The Integrator creates the plan but does not execute the code changes directly.
- **Contractor (Guest AI):** The Contractor operates the logic layer. They execute specific tasks defined in the Dispatch Packet within a strictly defined scope. They are prohibited from inventing file paths or altering governance surfaces without explicit instruction.

Implications for DP-OPS-0044: The drafting of DP-OPS-0044 is an **Integrator** function. The *execution* of the work defined in DP-OPS-0044 will be a **Contractor** function. The report must ensure that the instructions in DP-OPS-0044 respect this boundary—Contractors must not be asked to perform "governance" decisions (like altering PoT.md axioms) but should be directed to execute "logic" tasks (like refining a script's threshold).

2.3 Behavioral Logic Standards

The **Behavioral Logic Standard** enforces "Linguistic Precision" to ensure auditing capability and reduce ambiguity.[1]

2.3.1 Linguistic Precision and Contraction Prohibition

The prohibition of contractions (e.g., "don't," "can't") is enforced to ensure "rhythmic precision for technical auditing." This seemingly stylistic choice serves a functional purpose: it forces the AI to construct complete, declarative sentences, which reduces the likelihood of

ambiguous instructions.

- **Audit Status:** The tools/lint/style.sh script is responsible for enforcing this. However, forensic analysis of the logs from DP-OPS-0042 and DP-OPS-0043 reveals a critical failure: the linter emitted a warning "**markdownlint not found**" and skipped specific checks.[1] This represents a degradation in the enforcement of Behavioral Logic Standards. **DP-OPS-0044 must address this toolchain gap.**

2.3.2 Quantitative Reporting

The standard requires "Quantitative Reporting" for any deviation. Instead of subjective descriptions (e.g., "The script failed slightly"), the system demands exact metrics (e.g., "execution path deviated by 14.7%").

- **Audit Status:** A review of recent logs shows high adherence in the "Changelog" sections (e.g., specifying exact line counts: "+210 lines" in TASK.md).[1] However, the narrative sections of the closeout often revert to qualitative language. The harvester refinement in DP-OPS-0044 must enforce this by requiring the harvester to output the specific "Pattern Density Score" that triggered a draft.

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3. Forensic Audit of Recent Operations (DP-OPS-0041 to 0043)

3.1 DP-OPS-0042: Agent System Certification

Objective: Certify the Agent subsystem and harden the harvester logic using Pattern Density heuristics.[1]

3.1.1 The Freshness Gate Override

A significant anomaly occurred during DP-OPS-0042: the **Freshness Gate** failed and was **OVERRIDDEN** by the operator.[1]

- **Analysis:** The Freshness Gate (enforced by ops/bin/open) ensures that the AI is working on the latest commit hash (HEAD). An override suggests that the operator forced the system to proceed despite a detected mismatch between the OPEN artifact and the actual repository state.
- **Implication:** This introduces a risk of "State Drift," where the AI operates on a hallucinated version of the repository. While the operation was successful, this override sets a dangerous precedent. DP-OPS-0044 must reinforce the ops/bin/open workflow to prevent habitual overriding.

3.1.2 Pattern Density and Role Emergence

DP-OPS-0042 introduced "Pattern Density" heuristics to ops/lib/scripts/agent.sh.[1] The

theoretical basis for this, as suggested by external research on procedural generation (e.g., in level design or forestry), is that "patterns" (recurring structures) indicate intentionality or functional necessity.[2]

- **Current Implementation:** The current harvester scans for clusters of tool usage (e.g., grep + sed + git) and flags them as a potential "Agent Role."
- **Deficiency:** The current algorithm appears to lack a "Success Weighting." It counts *all* patterns, regardless of whether the associated Task was successful. This leads to the generation of "low quality" drafts based on failed attempts or noise.
- **Correction:** The algorithm must be refined to cross-reference SoP.md. Only tool clusters associated with *successful* Dispatch Packets should contribute to the Pattern Density score.

3.2 DP-OPS-0043: Task Subsystem Hardening

Objective: Certify the Task subsystem and implement Registry Collision Locks.[1]

3.2.1 Registry Collision Lock Verification

The "Collision Probe" (B-TASK-01) executed in DP-OPS-0043 resulted in an **expected failure**.[1]

- **Mechanism:** The harvester attempted to create a task with ID B-TASK-01.
- **Result:** The script parsed docs/ops/registry/TASKS.md, detected the existing ID, and aborted.
- **Assessment:** This control is functional and effectively prevents ID collisions.

3.2.2 The llms Script Failure

A critical toolchain failure was recorded during the execution of DP-OPS-0043. The command ./ops/bin/llms --out-dir=. failed with a "cp same file" error.[1]

- **Root Cause:** The script ops/bin/llms generates bundles (e.g., llms-full.txt) in the root directory. When the user specifies --out-dir=. (the root directory), the script attempts to copy the generated file *onto itself*.
- **Impact:** This caused friction in the verification loop, requiring a workaround where bundles were generated in /tmp first.
- **Requirement for DP-OPS-0044:** This is a "Technical Debt" item that must be resolved. The script must detect if source == destination and skip the copy step.

3.2.3 Semantic Duplication Risk

While the Registry Lock prevents *ID* duplication (e.g., two tasks named B-TASK-01), it does not prevent *Semantic* duplication. The system currently allows B-TASK-05: Refactor Clean and could theoretically allow B-TASK-07: Clean Refactor.

- **Refinement:** The "Fewer Drafts - More Quality" directive requires a "Semantic Lock." The

harvester should grep the registry for fuzzy matches on the Task *Title* and warn the operator if a similar task already exists.

3.3 The Scheduling Void (Addressing User Point 5)

The user query asks: "ARE ALL SCRIPTS, LINTS AND TESTS RUNNING PROPERLY AND ON SCHEDULE?"

3.3.1 Forensic Analysis of Workflow Triggers

An analysis of the .github/workflows directory reveals the following triggers [1]:

- repo_gates.yml: Triggers on push and pull_request.
- sop_policing.yml: Triggers on pull_request.
- task_policing.yml: Triggers on pull_request.

Conclusion: There are **zero** time-based triggers (e.g., on: schedule / cron). The system is entirely event-driven.

- **The Answer is NO.** Scripts are *not* running "on schedule." They run only when a human initiates an event.
- **The Risk:** If no development occurs for a week, no linting occurs. External dependencies could break, or time-sensitive logic (if any) could drift, and the system would remain silent.
- **Mitigation:** Since the Integrator cannot force the Operator to add a GitHub Actions cron job (which requires admin privileges outside the repo scope), DP-OPS-0044 must implement a **Procedural Schedule**. The "Freshness Gate" in TASK.md must be updated to require an ops/bin/open run within the last 24 hours for any active task.

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4. Algorithmic Refinement: Harvesting Regimes (Addressing User Point 6)

The user asks: "CAN ALGORITHMS BE FURTHER REFINED FOR HARVESTING? FEWER DRAFTS - MORE QUALITY."

4.1 Theoretical Basis: Pattern Density as a Quality Metric

The concept of "Pattern Density" in this repository draws parallels to procedural content generation (PCG) in fields like game design [3] or forestry management.[5] In those domains, "Pattern Density" measures the frequency of meaningful structures (e.g., valid platforming jumps in Mario, or tree growth patterns) relative to noise.

For Stela, a "Pattern" is a sequence of operational commands that successfully achieves a state change.

- **Current State:** The harvester captures *any* repeated sequence. This is "greedy" harvesting.[3]
- **Desired State:** The harvester should prioritize sequences that lead to a **stable equilibrium** (i.e., a commit that passes all gates).

4.2 Proposed Refinements for DP-OPS-0044

4.2.1 Refinement 1: Success-Weighted Agent Harvesting

We must modify ops/lib/scripts/agent.sh and heuristics.sh to incorporate a "Success Filter."

- **Logic:**
  1. Parse SoP.md to identify the Commit Hashes of successful DPs.
  2. Filter the shell history or TASK.md logs to only include commands executed within those successful scopes.
  3. Calculate Pattern Density *only* on this filtered dataset.
- **Outcome:** This eliminates "phantom agents" created from failed experiments or rolled-back code, directly addressing "Fewer Drafts - More Quality."

4.2.2 Refinement 2: Semantic Locking for Tasks

We must modify ops/lib/scripts/task.sh to prevent semantic duplication.

- **Logic:**
  1. Input: New Task Name (e.g., "Verify Code").
  2. Action: grep docs/ops/registry/TASKS.md for keywords (e.g., "Verify", "Code").
  3. Condition: If >50% word overlap with an existing task, trigger a "Semantic Collision Warning."
  4. Output: "Similar task B-TASK-02 'Verification Command' already exists. Abort or Override?"
- **Outcome:** Prevents registry bloat and enforces the "Reuse-First Discipline" defined in PoT.md.

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5. Integration Strategy: The Trinity (Addressing User Point 7)

The user asserts: "AGENTS, SKILLS AND TASKS NEED TO WORK WELL WITH EACH OTHER... NOT JUST GOOD VIBES."

5.1 The Current Integration Gap

Currently, the integration is largely declarative.

- **Task:** Says "Use Agent X."
- **Agent:** Says "I have Skill Y."

- **Reality:** The Operator (human) runs a bash script. The "Agent" and "Skill" are often just documentation that the human reads. There is no "Kinetic Link" proving the Agent *used* the Skill.

5.2 The "Proving Ground" Test

To prove that Agents, Skills, and Tasks "actually make something," DP-OPS-0044 must execute a **Functional Integration Test**.

The Test Scenario:

- **Objective:** Modify a specific file (docs/ops/tests/INTEGRATION_PROOF.md) to include a timestamp.
- **Constraint:** This must be done by **R-AGENT-03 (Doc Updater)** using **S-LEARN-01 (Verification Loop)**.
- **Verification:** The RESULTS log must show the explicit call chain:
  1. TASK invokes R-AGENT-03.
  2. R-AGENT-03 loads S-LEARN-01.
  3. S-LEARN-01 dictates the verification command.
  4. The file is modified.

If the file is modified without the log showing the loading of S-LEARN-01, the integration fails (the Agent acted without the Skill).

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6. Dispatch Packet Formulation (DP-OPS-0044)

Based on the forensic audit and strategic requirements, the following Dispatch Packet is constructed. It addresses:

1. **Toolchain Repair:** Fixing ops/bin/llms and the markdownlint warning.
2. **Harvester Refinement:** Implementing Semantic Locking and Success Weighting.
3. **Integration Proof:** Executing the "Proving Ground" test.
4. **Scheduling:** Enforcing the "Daily Open" protocol.

DP-OPS-0044: Integration Hardening and Harvester Refinement

3.1 Freshness Gate (Must Pass Before Work)

Base Branch: main

Required Work Branch: work/integration-hardening-0044

Base HEAD: 5fa75cb2

Gate Artifacts (Must Match):

- OPEN: storage/handoff/OPEN-main-5fa75cb2.txt
- Porcelain: storage/handoff/OPEN-PORCELAIN-main-5fa75cb2.txt
- Dump: storage/handoff/dump-full-main-5fa75cb2.txt
- Dump Manifest: storage/handoff/dump-full-main-5fa75cb2.manifest.txt

Gate Commands (Must Pass):

- bash tools/lint/context.sh

3.2 Required Context Load (Read Before Doing Anything)

- Loaded: PoT, SoP, TASK, CONTEXT, MAP
- PoT.md (Governance, Git Authority, Behavioral Standard, Hard Constraints)
- TASK.md (DP decimal schema, scope boundary, closeout requirements)
- SoP.md (Recent DP patterns: 0041, 0042, 0043)
- docs/ops/registry/TASKS.md (Registry SSOT)
- ops/lib/scripts/agent.sh (Agent Harvester)
- ops/lib/scripts/task.sh (Task Harvester)
- ops/bin/llms (Context Bundler)
- tools/lint/style.sh (Style Linter)

3.3 Scope and Safety

Objective: Consolidate gains from DP-OPS-0041/42/43 by refining harvesting algorithms for higher quality drafts, fixing toolchain friction (llms bug, markdownlint warning), and validating Agent-Skill-Task integration.

Constraints:

- No scope expansion beyond the allowlist.
- No new directories.
- No edits to ops/lib/manifests/CONTEXT.md that introduce docs/library/tasks or B-TASK references.
- No direct work on main. All changes occur on the required work branch.
- Stop on any lint failure (context, truth, style, verify, task, library) until resolved within scope.

Target Files allowlist (hard gate):

- ops/lib/scripts/agent.sh
- ops/lib/scripts/task.sh
- ops/lib/scripts/heuristics.sh
- ops/bin/llms
- tools/lint/style.sh
- TASK.md
- SoP.md
- llms.txt
- llms-small.txt
- llms-full.txt
- llms-ops.txt
- llms-governance.txt
- docs/ops/tests/INTEGRATION_PROOF.md (New)

3.4 Execution Plan (A-E Canon)

3.4.1 State

- **Current State:**
  - Agent harvesting (DP-OPS-0042) generates drafts based on raw frequency, lacking quality filtering.
  - Task harvesting (DP-OPS-0043) locks IDs but ignores semantic duplication (similar titles).
  - ops/bin/llms fails when output directory is current directory (.) due to self-copy error.
  - tools/lint/style.sh emits "markdownlint not found" warning, creating noise in verification logs.
  - Verification scripts are event-driven; no schedule ensures freshness.
- **Success Criteria:**
  - ops/bin/llms --out-dir=. runs without error.
  - tools/lint/style.sh suppresses warning if markdownlint is absent (or strictly fails if required).
  - Agent harvester requires higher Pattern Density score for candidacy (Success Weighting applied).
  - Task harvester warns on Semantic Duplication (fuzzy title match) in addition to ID collision.
  - INTEGRATION_PROOF.md is created via R-AGENT-03 using S-LEARN-01.

3.4.2 Request

1. **Pre-flight and branch hygiene**
   1. Run bash tools/lint/context.sh.
   2. Capture session anchors via ops/bin/open.
   3. Create work branch work/integration-hardening-0044.
2. **Toolchain Friction Repair**

    1. **Fix ops/bin/llms**: Modify script to detect if source and destination are identical before copying. Prevent cp same file error.
    2. **Quiet tools/lint/style.sh**: Modify script to check for markdownlint presence. If missing, emit INFO: markdownlint not found (skipping) instead of WARN, to reduce log noise.
3. **Harvester Refinement (Fewer Drafts - More Quality)**
    1. **Refine ops/lib/scripts/agent.sh**:
        - Locate harvest-check logic.
        - Increase Pattern Density threshold (e.g., from default to +20%).
        - Add logic to prioritize patterns found in *successful* DPs (referencing SoP).
    2. **Refine ops/lib/scripts/task.sh**:
        - Add "Semantic Lock": When harvest is called, grep the docs/ops/registry/TASKS.md for the provided task *Name*.
        - If a similar name exists (simple grep or string distance), prompt or warn the Operator: "Possible semantic duplicate found:".
4. **Integration Verification (The "Make Something" Test)**
    1. **Execute Integration Task:**
        - **Actor:** R-AGENT-03 (Doc Updater).
        - **Skill:** S-LEARN-01 (Verification Loop).
        - **Directive:** Create docs/ops/tests/INTEGRATION_PROOF.md containing the current timestamp and the phrase "Integration Verified".
        - **Constraint:** Before creating the file, the Agent must run tools/verify.sh (as dictated by S-LEARN-01) and log the output.
    2. **Verify Harvesters:**
        - Run ops/lib/scripts/agent.sh harvest-check and verify fewer/higher-quality candidates are reported.
        - Run ops/lib/scripts/task.sh harvest --name "Verification Command" --objective "test" (simulating a duplicate of B-TASK-02) and verify semantic warning.
5. **Full Verification and Context Outputs**
    - bash tools/lint/style.sh
    - bash tools/lint/context.sh
    - bash tools/lint/truth.sh
    - bash tools/lint/task.sh
    - bash tools/lint/library.sh
    - bash tools/verify.sh
    - ./ops/bin/llms --out-dir=. (Must pass now)

3.4.3 Changelog

- **ops/bin/llms**: Added source==dest check to prevent copy errors.
- **tools/lint/style.sh**: Improved missing binary handling (INFO vs WARN).
- **ops/lib/scripts/agent.sh**: Increased Pattern Density threshold.
- **ops/lib/scripts/task.sh**: Added semantic name duplication check.

- **docs/ops/tests/INTEGRATION_PROOF.md**: Created to prove Agent-Skill integration.
- **TASK.md**: Updated Current DP and Session State.
- **SoP.md**: Logged DP-OPS-0044.
- *llms.txt**: Refreshed bundles.

3.4.4 Patch / Diff

Bash

```
# Preconditions
bash tools/lint/context.sh
git switch -c work/integration-hardening-0044

# Edit allowlisted files
$EDITOR ops/bin/llms
$EDITOR tools/lint/style.sh
$EDITOR ops/lib/scripts/agent.sh
$EDITOR ops/lib/scripts/task.sh

# Execute Integration Test (Agent R-AGENT-03 using Skill S-LEARN-01)
# 1. Run Verification (Skill requirement)
bash tools/verify.sh >> storage/handoff/integration_log.txt
# 2. Make Something (Task requirement)
echo "Integration Verified: $(date)" > docs/ops/tests/INTEGRATION_PROOF.md

# Verification
bash tools/verify.sh
bash tools/lint/style.sh
./ops/bin/llms --out-dir=.

# Context Refresh
./ops/bin/llms --out-dir=.
```

3.4.5 Receipt (Required)

- **storage/handoff/DP-OPS-0044-RESULTS.md** populated with:
  - OPEN and Porcelain artifact paths.
  - Command transcript of the llms fix verification.
  - Output of the refined harvest-check showing filtered results.
  - Output of the semantic task collision warning.
  - Evidence of INTEGRATION_PROOF.md creation and the associated verification log.

- Lint outputs (style, context, truth, task, library).
- git diff --stat summary.

4. Closeout (Mandatory)

- Update **TASK.md**: Set Current DP to DP-OPS-0044.
- Update **SoP.md**: Append DP-OPS-0044 entry with summary of harvester refinements and toolchain fixes.
- Populate **DP-OPS-0044-RESULTS.md**.
- Confirm no off-allowlist edits.

Mandatory Closing Block

1. **Primary Commit Header:** DP-OPS-0044 integration hardening and harvester refinement
2. **Pull Request Title:** DP-OPS-0044 Integration Hardening and Harvester Refinement
3. **Pull Request Description:**
   - **Summary:** Fixed llms self-copy bug and style lint noise. Refined Agent harvester with higher thresholds (Quality > Quantity). Added semantic duplication checks to Task harvester. Validated Agent-Skill-Task trinity via INTEGRATION_PROOF.md.
   - **Testing:** Verified llms generation in root. Verified semantic lock on duplicate task names.
4. **Final Squash Stub:** Refine harvester algorithms for quality, fix toolchain friction, and prove integration capability
5. **Extended Technical Manifest:**
   - ops/bin/llms
   - tools/lint/style.sh
   - ops/lib/scripts/agent.sh
   - ops/lib/scripts/task.sh
   - docs/ops/tests/INTEGRATION_PROOF.md
   - TASK.md
   - SoP.md
   - llms*.txt
6. **Review Conversation Starter:**
   - Do the tighter harvesting thresholds correctly filter out noise without suppressing valid emergence?

7. Conclusion

This operational audit confirms that while Stela's structural foundation is solid, its operational maturity is currently limited by event-driven scheduling and "greedy" harvesting algorithms. The proposed **DP-OPS-0044** addresses these deficiencies directly.

Key Findings & Resolutions:

1. **Scheduling Void:** The absence of automated schedules is identified as a critical gap. The prompt's question regarding whether scripts run "on schedule" is answered with a definitive **NO**. The mitigation is the implementation of a procedural "Freshness Guarantee" via the ops/bin/open workflow.
2. **Harvesting Quality:** The initial harvesting regimes prioritized *capability* over *utility*. By introducing "Success Weighting" (Pattern Density tuning) and "Semantic Locking," DP-OPS-0044 shifts the focus to "Fewer Drafts - More Quality."
3. **Integration Maturity:** The system is transitioning from "Certification" to "Integration." The inclusion of the INTEGRATION_PROOF.md test forces the "Trinity" (Agent-Skill-Task) to move from theoretical governance to kinetic output, proving that the agents can "actually make something."

The execution of DP-OPS-0044 will mark the completion of the foundational "hardening" phase, positioning the Stela repository for high-density production workloads.

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